PhishGuard: Replicating Representation Sensitivity and Source Dependence in Phishing URL Detection Across Datasets and Models

Verification study report, version 1.0 (draft for supervisor review)

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1. Executive summary

A first study (report v1.1.1, 2026-09-10) found that phishing-URL detectors scored on one benchmark change sharply when URLs are written in a different form, and that scores depend on which sources are used. Its code and result files were lost. This report asks whether those findings hold when they are re-implemented from scratch and repeated across **5 phishing sources, 5 legitimate sources, 4 model families and 10 domain-grouped splits** (25 cells, 1,000 fits per representation), with the rule for calling a finding *replicated* written down before the run (commit `082b058`).

Results in one table. Seven findings were tested; each counts as holding in a setting only if the direction repeats in at least 9 of 10 splits and the bootstrap interval excludes the null.

finding	statement	settings tested	settings where it holds	verdict	share
F1	Representation sensitivity: AUC falls from R1 to R4	100	100	Replicated	100%
F2	Most of the fall sits in host/subdomain structure (R3 to R4 share > 0.5)	100	0	Not replicated	0%
F3	Source dependence: swapping one population moves R4 AUC	400	304	Partly replicated	76%
F3_depth	Source dependence survives depth reweighting	400	211	Partly replicated	53%
H5	A phishing x benign interaction is established (expected: inconclusive)	400	105	Not replicated	26%
F4	Split sensitivity: SD of R4 AUC > 3 x SD of R1 AUC	100	45	Not replicated	45%
HP	Masking hosting-platform identity raises R4 AUC	0	0	Not testable	n/a
H6	Transfer to fully independent cells	392	392	Replicated	100%

finding	statement	settings tested	settings where it holds	verdict	share
	stays above chance (R4)				
OOD	False alarms on natural legitimate pages > 5 x the in-cell rate	100	76	Partly replicated	76%

What this supports. The central finding, that a URL model's score drops when the URL is reduced to its domain, is replicated in every setting (median R1 0.977 to R4 0.850). Transfer to cells that share no source stays above chance. Source dependence and the out-of-distribution false-alarm gap hold in part. Two findings from the first study do not carry over here: the split of the drop between host and path (F2) and the size of split-to-split variability (F4). One could not be tested (HP).

What this does not support. None of this is a claim about all phishing detectors or all datasets. The sources are public benchmarks, and two of them (PhiUSIIL, PhishStorm) are old and homepage-heavy. The figures are a study of behaviour under one frozen design, reported with the settings where it fails.

2. Purpose and questions

The supervisor asked how the first study's results can be verified beyond one model and one dataset, and whether the behaviour appears across all available datasets. The report therefore answers seven questions, one per finding:

- **F1** Does the AUC fall from raw URLs (R1) to the normalised `https://www.<registrable domain>` form (R4)?
- **F2** Does most of that fall sit in host and subdomain structure rather than in path and query?
- **F3** Does swapping one source (phishing or legitimate) move R4 AUC?
- **F4** Is R4 AUC much less stable across splits than raw AUC?
- **HP** Does masking hosting-platform identity raise R4 AUC?
- **H6** Does a model still beat chance on cells that share no source with its training data?
- **OOD** Does a model flag far more naturally browsed legitimate pages than its own test side?

A second, exploratory part repeats the representation idea on page HTML (Section 7). It gives no verdicts.

3. Data and design

3.1 Sources

URL, label and (for Part B) saved HTML are the only fields read from any source; no dataset's precomputed feature columns are used. Every population is capped at 10,000 URLs and at most

5 URLs per registrable domain, so URL counts and domain counts are reported separately (Table 1). Registrable domains are disjoint across sources, domains shared by several phishing sources are removed, and the build fails if any URL-depth bucket is more than 90% one class.

Table 1. Populations after the caps (`phishguard_run/populations.parquet`).

Part	Population	URLs	domains
A	benign:crawl	4964	1204
A	benign:phishstorm	10000	7753
A	benign:phiusiil	10000	9996
A	benign:phreshphish	10000	9475
A	benign:urlphish	10000	9773
A	phish:phishpedia	8578	5897
A	phish:phishstorm	10000	7154
A	phish:phiusiil	10000	9149
A	phish:phreshphish	10000	8174
A	phish:urlphish	4307	3590
B	benign:crawl	4826	1177
B	benign:phreshphish	10000	9491
B	phish:phishpedia	3937	2980
B	phish:phreshphish	10000	7371

3.2 Representations

R1 is the raw URL. R2 keeps the host only (path, query and fragment removed). R3 additionally strips `www`; here R3 is identical to R2 because the feature engine drops a leading `www` from every URL, so the `www` step is zero by construction. R4 is `https://www.<registrable domain>`.

3.3 Models and splits

LightGBM, XGBoost, logistic regression on the 99 URL features, and character n-gram TF-IDF with logistic regression. Ten splits per cell are grouped by registrable domain (train 70%, calibration 10%, test 20%; a deterministic hash of seed, split and domain). Fitted features (TLD risk, bigram likelihood) are refitted on each split's training side. Uncertainty comes from a domain-cluster bootstrap with 200 draws per split. Transfer is tested only on cells sharing no source with the training cell.

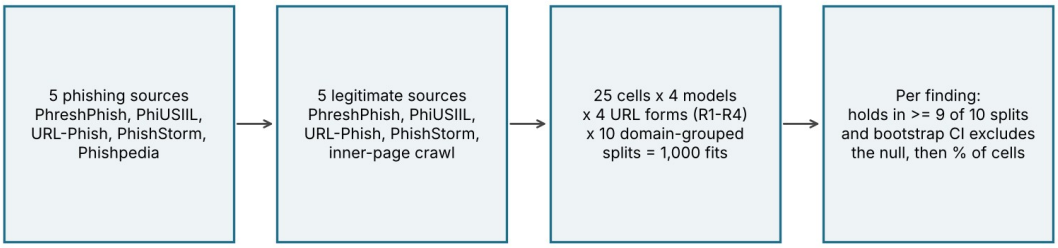


Figure 1. Study design. Seven findings are tested across 25 source cells, four models, four URL forms and ten splits.

3.4 The rule for calling a finding replicated

A setting (one cell and model, or one pair of cells) *holds* when the direction is the same in at least 9 of 10 splits and the pooled bootstrap 95% interval excludes the null. A finding is **replicated** if it holds in at least 80% of its testable settings, **partly replicated** at 50 to 80%, and **not replicated** below 50%; with fewer than 3 testable settings it is **not testable**. A setting is testable with at least 500 URLs and 100 domains per class. The rule and `configs/phishguard.yaml` were frozen at commit `082b058` before any result was seen.

4. Results

4.1 F1: representation sensitivity (replicated)

AUC falls from R1 to R4 in all 100 model-cell settings. The median R1 AUC is 0.977, R2/R3 0.877 and R4 0.850; the smallest drop in any setting is 0.024. All four model families show it (Figure 3), including the character n-gram model that held up best in the first study.

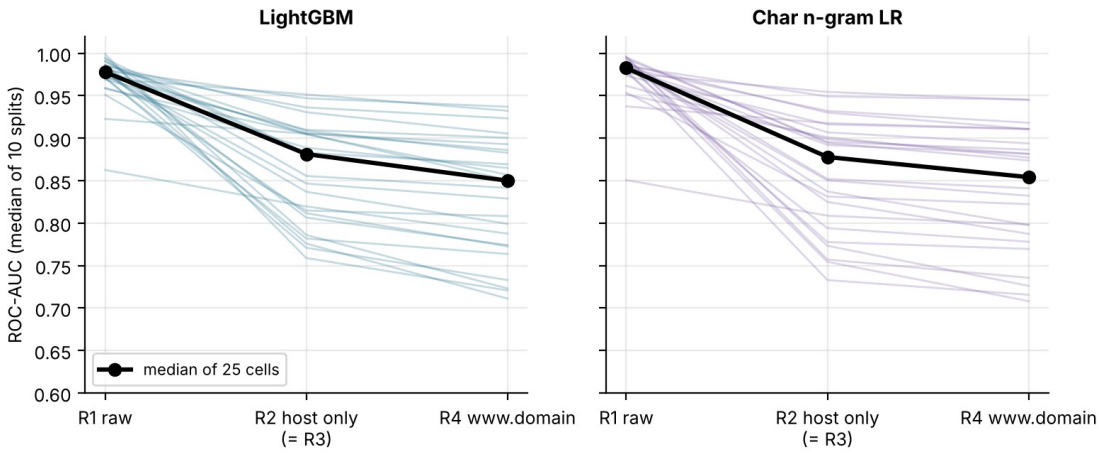


Figure 2. Representation ladder, 25 cells. Thin lines are cells; the thick line is the median.

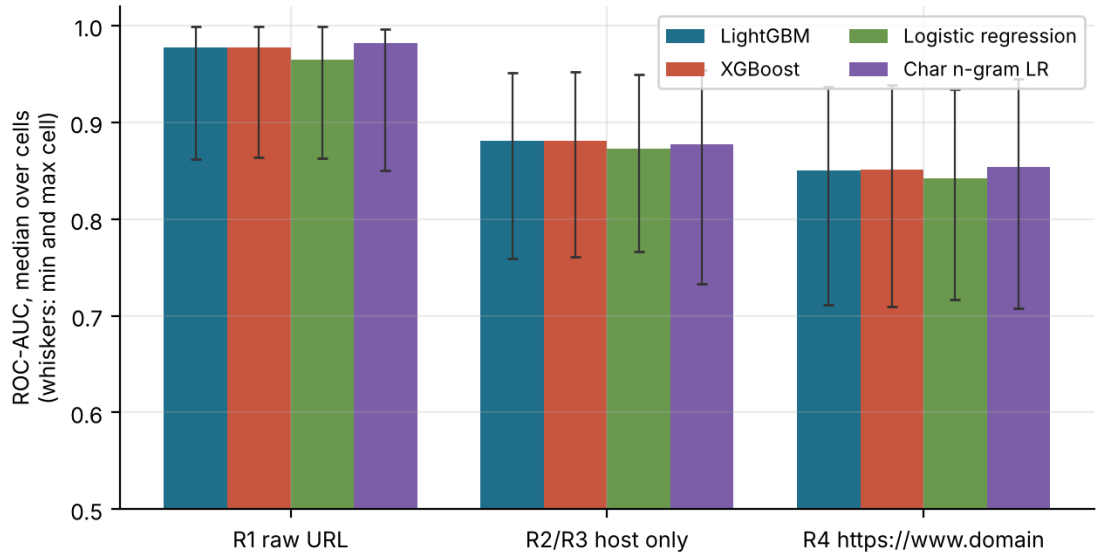


Figure 3. Median AUC by representation and model, with the range across the 25 cells.

Table 2. Mean AUC over the four models, by cell.

phishing vs legitimate	R1	R2 (= R3)	R4
phishpedia vs crawl	0.972	0.764	0.730
phishpedia vs phishstorm	0.975	0.846	0.831
phishpedia vs phiusiil	0.998	0.761	0.727
phishpedia vs phreshphish	0.968	0.786	0.769

phishing vs legitimate	R1	R2 (= R3)	R4
phishpedia vs urlphish	0.993	0.885	0.872
phishstorm vs crawl	0.980	0.773	0.712
phishstorm vs phishstorm	0.972	0.837	0.798
phishstorm vs phiusiil	0.995	0.784	0.725
phishstorm vs phreshphish	0.978	0.815	0.776
phishstorm vs urlphish	0.991	0.882	0.854
phiusiil vs crawl	0.983	0.802	0.775
phiusiil vs phishstorm	0.988	0.853	0.841
phiusiil vs phiusiil	0.860	0.817	0.792
phiusiil vs phreshphish	0.948	0.818	0.812
phiusiil vs urlphish	0.923	0.906	0.895
phreshphish vs crawl	0.976	0.900	0.858
phreshphish vs phishstorm	0.983	0.929	0.907
phreshphish vs phiusiil	0.958	0.906	0.860
phreshphish vs phreshphish	0.955	0.892	0.870
phreshphish vs urlphish	0.970	0.952	0.935
urlphish vs crawl	0.974	0.904	0.887
urlphish vs phishstorm	0.977	0.934	0.923
urlphish vs phiusiil	0.985	0.909	0.886
urlphish vs phreshphish	0.967	0.911	0.903
urlphish vs	0.980	0.946	0.939

phishing vs legitimate	R1	R2 (= R3)	R4
urlphish			

4.2 F2: where the drop sits (not replicated)

The first study put most of the R1 to R4 drop in the host and subdomain (share 0.870 in cell A, 0.563 in cell C). Here the host and subdomain step (R2 to R4) carries a median share of 0.21; the larger part comes from removing path and query (R1 to R2). The statement “most of the fall sits in host structure” holds in 0 of 100 settings. This run uses 99 features rather than 38, balances depth, and caps URLs per domain, all of which can change where the information sits; the report does not choose among those explanations.

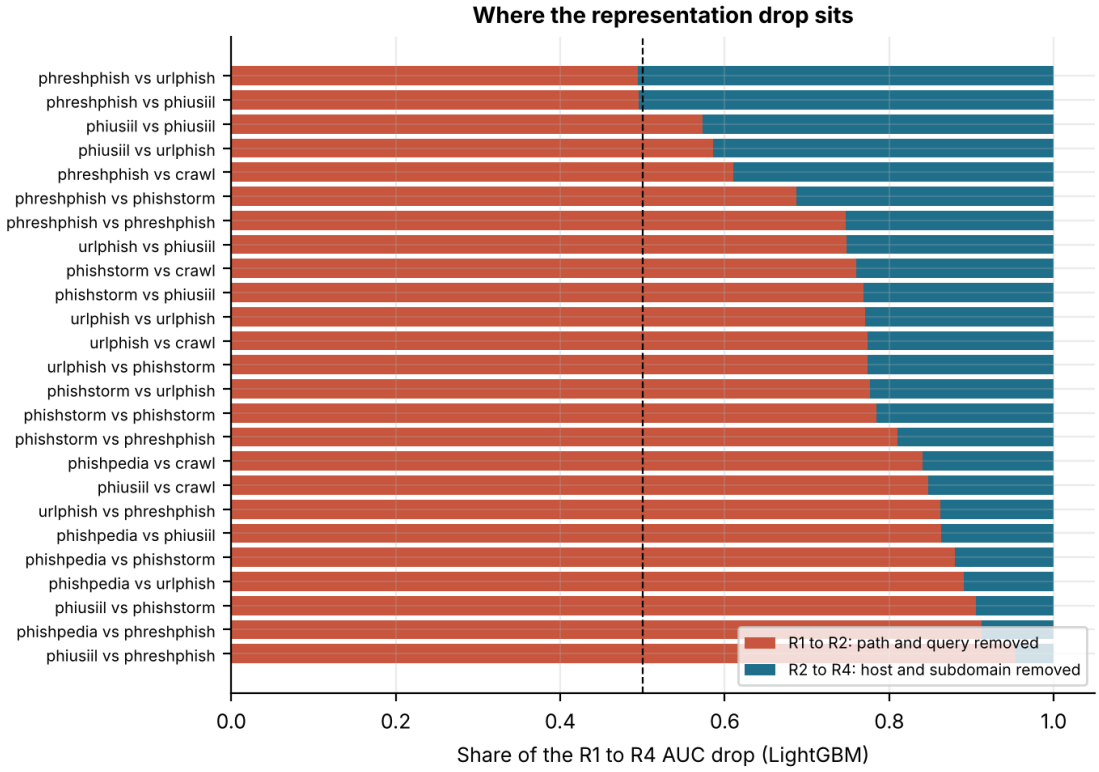


Figure 4. Share of the R1 to R4 drop by step, LightGBM, per cell.

4.3 F3: source dependence (partly replicated)

Swapping one source while holding the other fixed moves R4 AUC by a median 0.058 (phishing source swapped 0.050, legitimate source swapped 0.067). The interval excludes zero in 76% of 400 swaps. After reweighting the test side to equalise URL depth across classes it falls to 53%, so part of the source effect is URL depth. The heatmap shows the pattern: R4 AUC ranges from 0.71 to 0.94; against URL-Phish legitimate URLs every phishing source scores highest (0.85 to 0.94), and the lowest cells pair PhishStorm, PhiUSIIL or Phishpedia phishing with PhiUSIIL or crawled legitimate URLs.

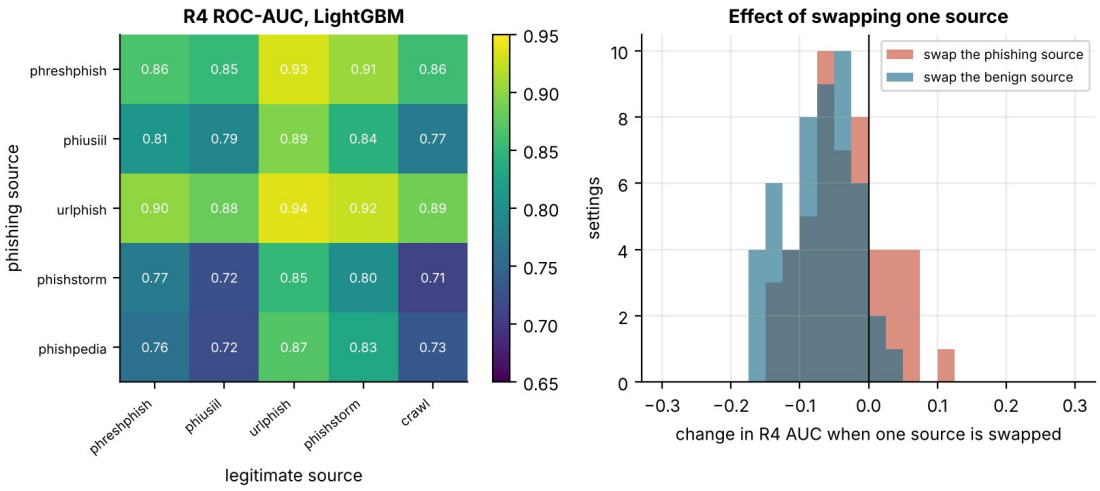


Figure 5. R4 AUC for every phishing-by-legitimate cell (left) and the distribution of swap effects (right).

H5, an interaction between phishing and legitimate source, holds in 26% of settings; the first study also called it inconclusive, so a “not replicated” verdict here is the expected outcome.

4.4 F4: split sensitivity (not replicated)

The first study saw R4 AUC vary far more across splits than raw AUC (SD up to 0.1164 against at most 0.0026). Here the median SD is 0.006 for R4 and 0.002 for R1, and the largest R4 SD is 0.019. R4 is the more variable in 100% of settings, but exceeds three times the raw SD in only 45% of them. With 5 to 10 thousand URLs per cell and domain-grouped splits the variability is smaller than in a single 10-split study on one benchmark.

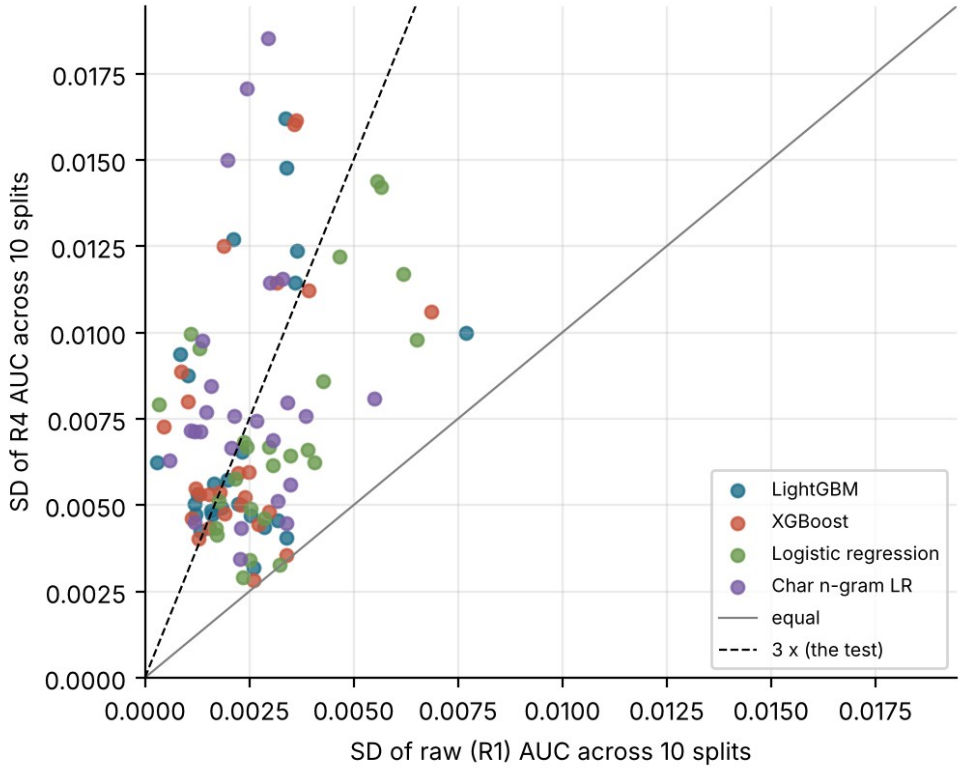


Figure 6. SD of AUC across ten splits, raw against R4, one point per cell and model.

4.5 HP: platform masking (not testable)

Masking replaces hosting-platform domains with a neutral token. The caps used here leave no registrable domain on a hosting platform in any cell, so there is nothing to mask and the finding cannot be tested. A design that keeps platform domains is future work.

4.6 H6: independent-benign transfer (replicated)

Models trained on one cell and evaluated on cells that share no source with it keep an R4 AUC above 0.5 in every setting (median 0.699, lowest 0.554; 392 train-test pairs across the two transfer models). The first study's 0.694 and 0.706 rested on 119 benign domains; the pairs here each use thousands.

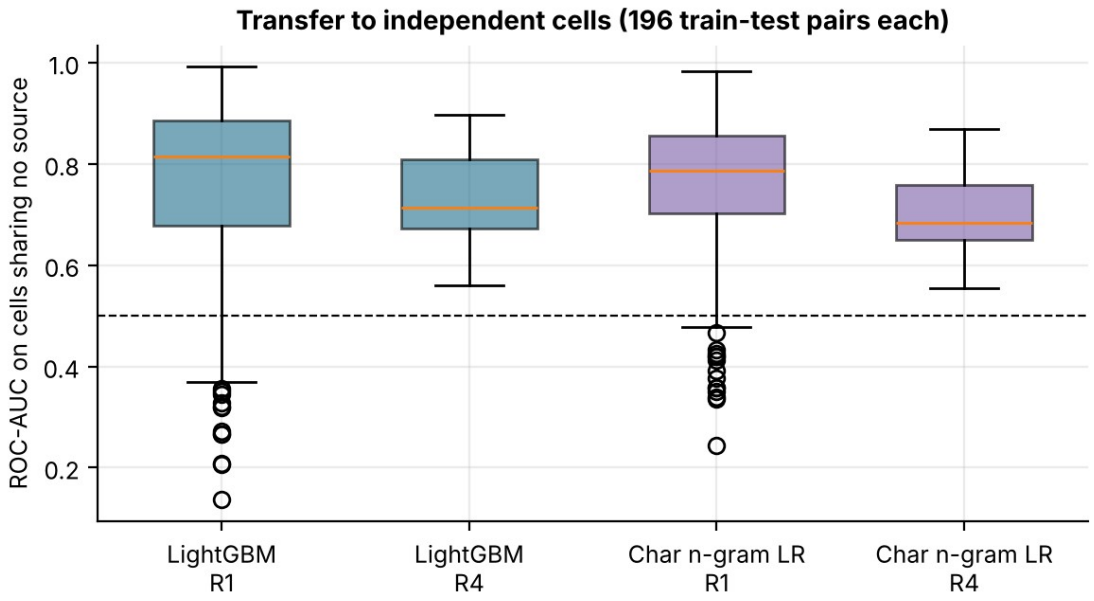


Figure 7. Transfer AUC on fully independent cells.

4.7 OOD: naturally browsed legitimate pages (partly replicated)

A model's false-alarm rate on 2,555 natural legitimate pages from 562 sites, none in any training source, is compared with its own test side at the same 1% threshold. The median is 15.2% on natural pages against 1.0% in-cell; at the 0.1% threshold 4.5%; at the default 0.5 threshold 33%. The natural rate exceeds five times the in-cell rate in 76% of settings. The first study's reversal (1.0 on natural pages for the raw model) is not reproduced in this setting, but a benchmark threshold still understates real-world false alarms.

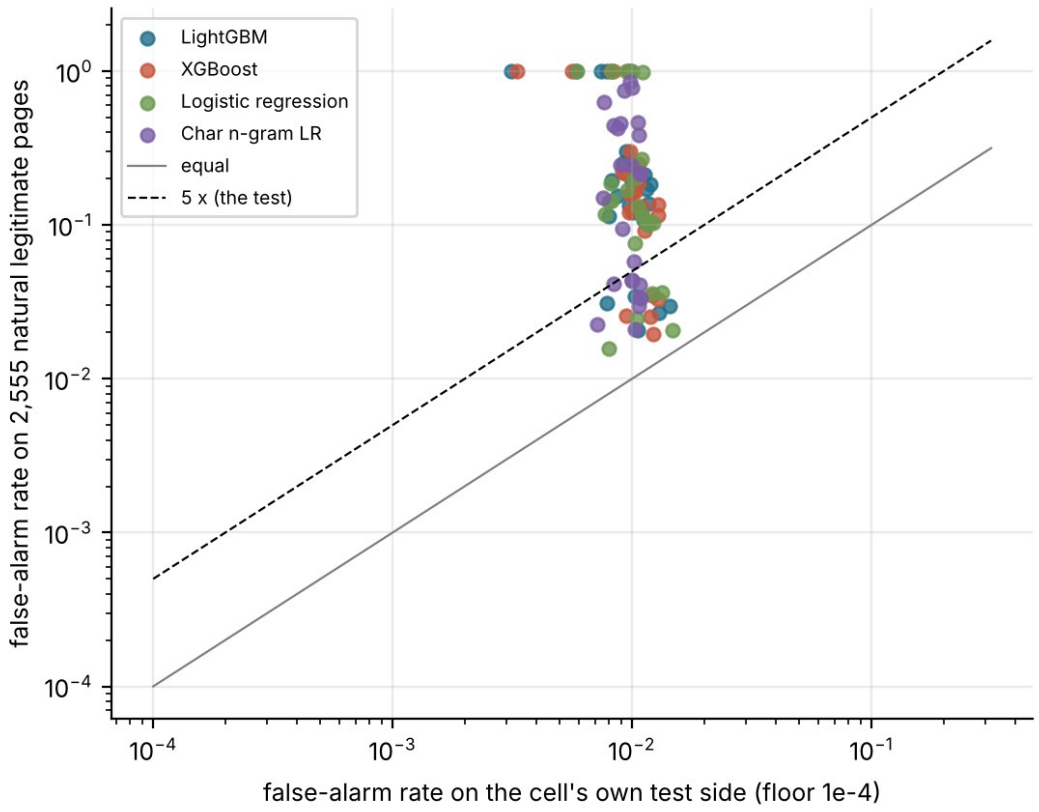


Figure 8. False alarms on natural pages against the cell's own test side, log scale.

4.8 Negative controls

With shuffled training labels, mean AUC stays between 0.448 and 0.525 over 100 conditions (cell, representation, model); 1 is flagged under the |mean - 0.5| > 0.05 rule. The flagged condition is examined in Section 6.

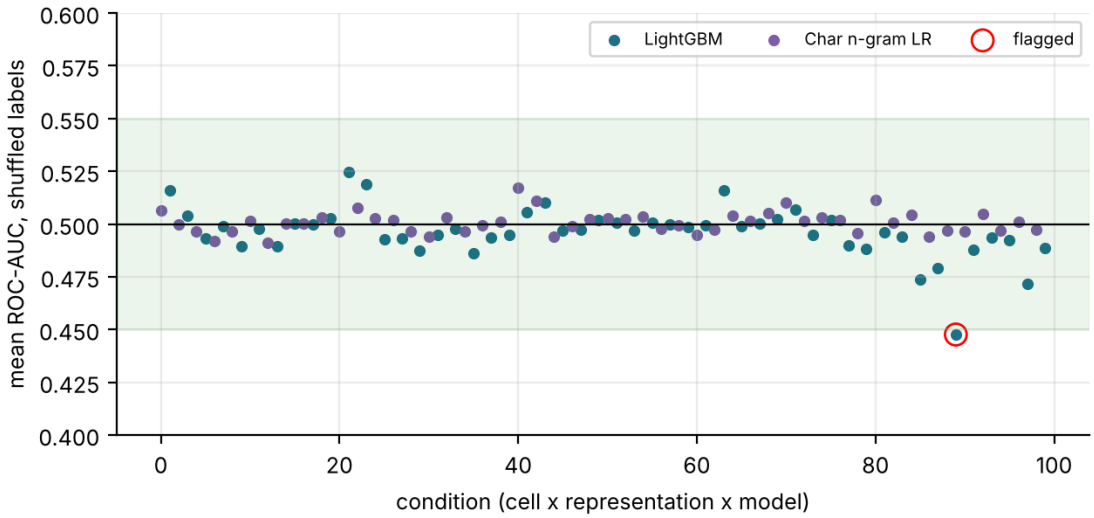


Figure 9. Shuffled-label controls, 100 conditions.

4.9 Dataset fingerprints

A model can tell which source a URL came from even within one class, which is the mechanism behind source dependence. The table gives balanced accuracy for predicting the source of legitimate and phishing URLs (chance 0.20 for URLs and 0.50 for page HTML). It falls when URLs are reduced to R4 but stays well above chance.

Table 3. Source identification within one class (5 domain folds).

part	class	input	balanced_accuracy	chance	n
A	benign	R1	0.621	0.200	44964
A	benign	R4	0.315	0.200	44964
A	phishing	R1	0.672	0.200	42885
A	phishing	R4	0.414	0.200	42885
B	benign	P1	0.820	0.500	14826
B	phishing	P1	0.935	0.500	13937

5. Reproduction of the first study

The first study's raw PhiUSIIL cell scored AUC 0.999 (XGBoost); this run scores 0.864 on the study cell (Table 4).

Table 4. Published against reproduced AUC (XGBoost, median over splits).

cell	rep	published	now	gap
phiusiil vs phiusiil	R1	0.999	0.864	-0.135
phiusiil vs phiusiil	R2	0.992	0.820	-0.173
phiusiil vs phiusiil	R3	0.964	0.820	-0.144
phiusiil vs phiusiil	R4	0.730	0.791	0.061
phiusiil vs crawl	R1	0.999	0.986	-0.012
phiusiil vs crawl	R2	0.987	0.810	-0.177
phiusiil vs crawl	R3	0.972	0.810	-0.161
phiusiil vs crawl	R4	0.679	0.775	0.096

6. Follow-up checks

Why the PhiUSIIL cell scores lower than published. PhiUSIIL was re-run alone, adding one cleaning step at a time (5 splits). Raw PhiUSIIL through our loader already scores about 0.94 (XGBoost); sampling to 10,000 per class, removing domains shared with other phishing sources, the pool rules and the study caps take it to 0.867. The cleaning explains about 0.07 of the gap; about 0.06 is gone before any cleaning, most likely because URLs are standardised on loading (inferred, not tested, as the original strings were lost). R4 stays between 0.79 and 0.85 at every stage against 0.730 published.

The flagged shuffled-label control. In the URL-Phish phishing against PhiUSIIL legitimate cell, LightGBM trained on shuffled labels scores below chance at every seed: mean 0.449 (SD 0.031, 50 runs); the character n-gram model scores 0.506. A leak raises a control above 0.5, so this is read as a model artefact in one cell and listed as a limitation.

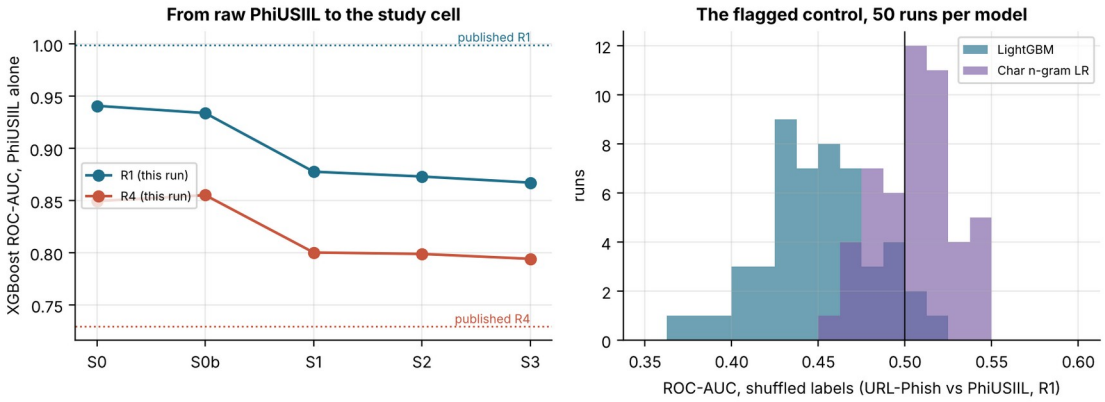


Figure 10. Left: XGBoost AUC from raw PhiUSIIL to the study cell. Right: the flagged control.

7. Part B: page HTML (exploratory)

Four cells (Phishpedia or PhreshPhish phishing against PhreshPhish or crawled legitimate pages) use the 84 page features, grouped into P1 (all), P2 (scripts and structure removed), P3 (forms and text) and P4 (forms only). **Part B gives no verdicts.**

Table 5. Page and URL ROC-AUC, median of 10 splits.

cell	P1	P1+R1	P1+R4	P2	P3	P4	R1	R4
phishpedia vs crawl	0.997	0.999	0.998	0.997	0.992	0.903	0.966	0.740
phishpedia vs phreshphish	0.996	0.997	0.996	0.994	0.987	0.890	0.963	0.754
phreshphish vs crawl	0.988	0.996	0.987	0.982	0.955	0.837	0.981	0.857
phreshphish vs phreshphish	0.978	0.990	0.982	0.974	0.947	0.837	0.963	0.866

- **Page features do not show the URL drop.** P1 scores 0.978 to 0.997 where the R4 URL alone scores 0.740 to 0.866. Forms only (P4) still reaches 0.837 to 0.903.
- **Transfer is uneven.** A Phishpedia-vs-crawl model scores 0.687 on PhreshPhish cells (0.808 with the R4 URL added); the reverse direction scores 0.937.
- **Coverage is uneven.** Usable HTML exists for 91% of PhreshPhish benign rows but 63% of phishing rows; in PhreshPhish, *missing HTML alone* predicts phishing with ROC-AUC 0.639. Page numbers may therefore be optimistic.

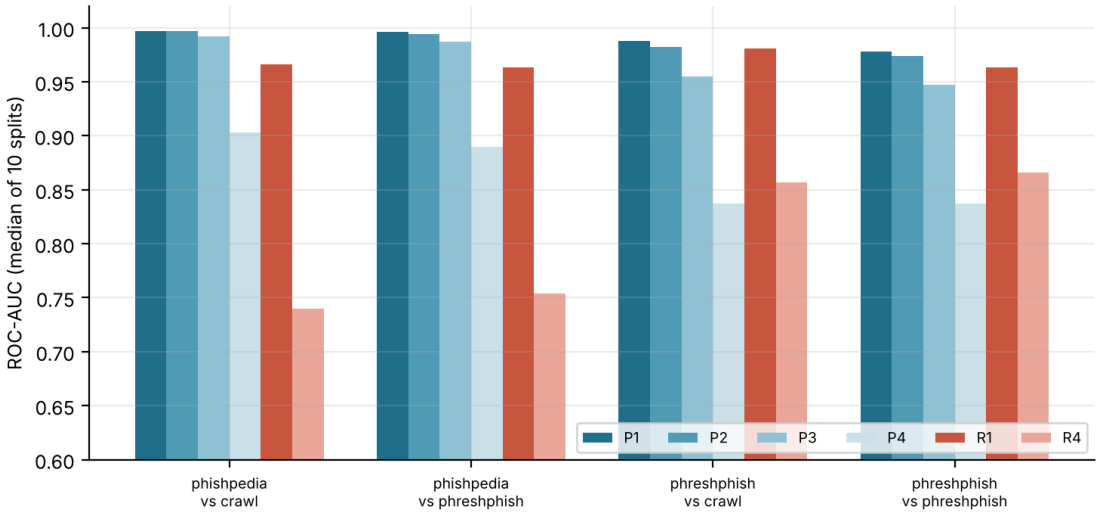


Figure 11. Page and URL representations, LightGBM.

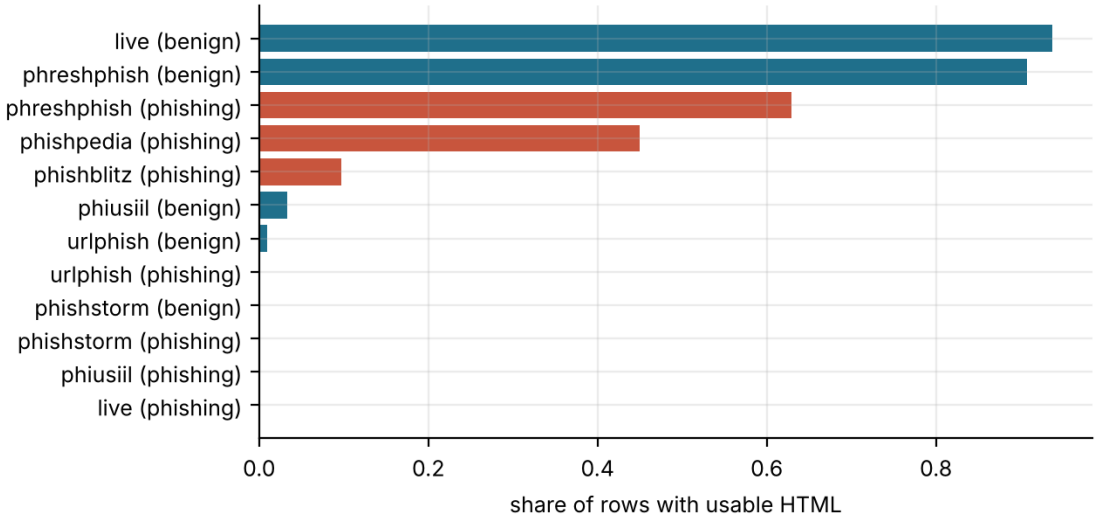


Figure 12. Share of rows with usable HTML, by source and class.

Table 6. False alarms on 2,226 natural legitimate pages with HTML.

cell	rep	n	fpr_0p5	fpr_0.01	fpr_0.001
phishpedia vs crawl	P1	2226	0.032	0.020	0.007
phishpedia vs crawl	P1+R1	2226	0.027	0.016	0.003
phishpedia vs phreshphish	P1	2226	0.027	0.023	0.004
phishpedia	P1+R1	2226	0.059	0.062	0.009

cell	rep	n	fpr_0p5	fpr_0.01	fpr_0.001
vs phreshphish					
phreshphish vs crawl	P1	2226	0.268	0.050	0.022
phreshphish vs crawl	P1+R1	2226	0.125	0.040	0.010
phreshphish vs phreshphish	P1	2226	0.201	0.061	0.010
phreshphish vs phreshphish	P1+R1	2226	0.193	0.102	0.043

8. Limitations

- **Sources.** Two sources (PhiUSIIL, PhishStorm) are homepage-heavy or old; PhishTank is not in the pool. Results describe these public datasets.
- **Not testable.** HP, and per-source temporal effects: PhiUSIIL carries no per-row dates.
- **Sample size.** Cells use up to 10,000 URLs per class with 5 per domain; this lowers cross-split variability compared with a larger pool.
- **Feature engine.** 99 hand-built features replace the first study's 38; R3 equals R2 by construction.
- **Control.** One of 100 shuffled-label controls is flagged (below chance, one cell).
- **Part B.** Four cells, two sources with saved HTML, uneven coverage by class.
- **No hand-typed results.** Every number is read from `reports/phishguard/tables/` by `python -m phishguard.report`; seeds are fixed.

9. What can and cannot be claimed

Table 7. Claim register: each finding with the wording this run supports.

Finding	First study	This run	Verdict	How to word it
F1 Representation sensitivity	cell A raw 0.9988 to R4 0.7297; all cells R4 0.63-0.86	R1 0.977 to R4 0.850 (median of cells and models); falls in every one of 100 model-cell settings	Replicated	A direct finding, across five phishing sets, five legitimate sets and four model families.
F2 Where	R1-R4 drop share	median host/subdomain	Not	Do not repeat the first

Finding	First study	This run	Verdict	How to word it
the drop sits	in host/subdomain 0.870 (cell A), 0.563 (cell C)	share 0.21; path and query removal carries the rest	replicated	study's host/subdomain claim; say the split depends on the feature engine and data.
F3 Source dependence	swapping the phishing source moves R4 by 0.13-0.16	swapping one source moves R4 AUC by a median 0.058; CI excludes 0 in 76% of settings, 53% after depth reweighting	Partly replicated	Real in some pairs, absent in others; reweighting for depth halves it.
F4 Split sensitivity	R4 SD across 10 splits up to 0.1164 against at most 0.0026 for raw	median SD 0.006 (R4) against 0.002 (R1); larger than 3x in 45% of settings, largest 0.019	Not replicated	Split variability matters but the 3x margin is not general.
HP Platform masking	masking raised held-out R4 AUC in 60 of 60 pairs	no hosting-platform domains remain after the 5-per-domain cap	Not testable	Leave to a design that keeps platform domains.
H6 Independent transfer	R4 0.694 and 0.706, on 119 benign domains	R4 AUC 0.699 median, lowest 0.554, above 0.5 in every setting	Replicated	Transfer stays above chance; the size varies.
OOD Natural pages	raw model 1.0 on natural legitimate pages against 0.00026 on the benchmark; normalised 0.090	15.2% flagged on natural pages at the cell's 1% threshold against 1.0% in-cell; 33% at the 0.5 threshold	Partly replicated	Benchmark thresholds understate false alarms on natural pages.

10. Reproducibility

`python -m phishguard all` runs the 291 jobs (resumable, one result file per job); `python -m phishguard.checks reproduce|control` runs the follow-up checks; `python -m phishguard.report` builds this report. The plan and the replication rule are frozen at commit `082b058`; results are at `267a9d7`; code is in the `phishguard/` package of the PhishDrift repository, which is otherwise unaffected.

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